

Geodesic complexity (it's actually quite simple)

Topological complexity [2] describes the difficulty of creating a continuous motion plan over a space X . X has topological complexity k (denoted $TC(X) = k$) if k is the minimal number such that there is an open cover $U_1 \cup U_2 \cup \dots \cup U_k = X \times X$ where each U_i has a local **motion planning rule**: a continuous map $\sigma_i : U_i \rightarrow PX$ (with PX the space of paths in X) such that $\sigma(x, y)$ is a path from x to y .

Geodesic complexity [4] is analogous to topological complexity, with the caveat that all paths are **geodesics** (shortest paths with constant speed). $GC(X) = k$ if k is the minimal number such that there is a **locally compact partition** $E_0 \sqcup E_1 \sqcup \dots \sqcup E_k = X \times X$ where each E_i has a local **Geodesic Motion Planning Rule (GMPR)** $\sigma_i : U_i \rightarrow GX$. Here, $GX \subseteq PX$ is the space of geodesics in X . We call $\sigma_0, \dots, \sigma_k$ a **geodesic motion planner**.

Theorem (simplex embedding theorem)

For integers $0 \leq k_0 < k$, if for each $i \in \{k_0, k_0 + 1, \dots, k\}$ there exists a non-empty collection $\mathcal{F}_i$ of embeddings into $X \times X$ such that:

- Each $\mathcal{F}_i$ consists of embeddings of i -dimensional simplices $\Delta_i \hookrightarrow X \times X$.
- For $F \in \mathcal{F}_i$, any geodesic in $\pi_{GX}^{-1}(\text{relint}^*(F))$ extends to a GMPR on all of $\text{img}(F)$.
- With $k_0 \leq i < k$, for any $F \in \mathcal{F}_i$ and GMPR Γ on $\text{img}(F)$, there is an $F' \in \mathcal{F}_{i+1}$ such that:
 - The map F is the restriction of F' to a face of Δ_{i+1} .
 - Each geodesic in $\Gamma(\text{img}(F))$ is separated from any geodesics in $\pi_{GX}^{-1}(\text{relint}^*(F'))$ by an open set.
 - There are a finite number of GMPRs on $\text{img}(F')$.

Then $GC(X) \geq k - k_0$ (i.e. any geodesic motion planner requires $\geq k - k_0 + 1$ sets).

Here, $\text{relint}^*(F)$ is the **relative interior** of $\text{img}(F)$, and $\pi_{GX} : GX \rightarrow X \times X$ maps a geodesic to its start and end point.

Proof sketch

This method is inspired by arguments of Davis [1] and Recio-Mitter [4], whose proofs use sequences, sequences of sequences, etc. of elements in $X \times X$. Each iteration is constructed so it must be in a unique GMPR. Our theorem describes sufficient conditions for constructing these sequences.

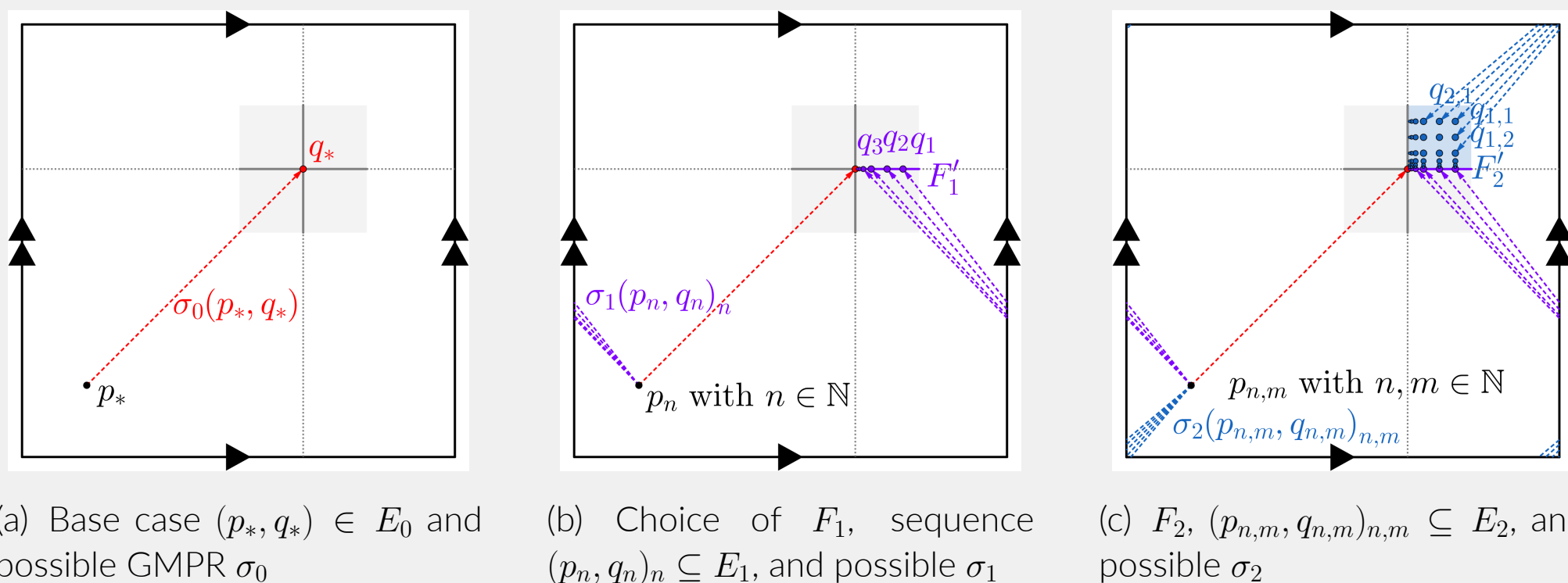


Figure 1. Mechanism of theorem when used on the 2-torus.

For X as the 2-torus, the sequence argument to show $GC(X) \geq 2$ behaves as follows: Assume for contradiction there is a geodesic motion planner σ_0, σ_1 . Let (p_*, q_*) be a point and its antipode, and assume $(p_*, q_*) \in E_0$. Construct $(p_n, q_n)_n \subseteq E_k$ where $p_i = p_*$, and q_i approaches q_* from the side opposite $\sigma_0(p_*, q_*)$ (as in Fig. 1(b)). Observe that σ_0 won't be continuous if $(p_n, q_n)_n \subseteq E_0$, so $(p_n, q_n)_n \subseteq E_1$. Repeat this process to produce $(p_{n,m}, q_{n,m})_{n,m} \subseteq E_k$ (as in Fig. 1(c)), and observe for both $k = 0$ and $k = 1$, σ_k must be discontinuous.

To prove our theorem, we construct sequences on the images of appropriately selected simplex embeddings ($\text{img}(F_1) = (p_*, F'_1)$, and $\text{img}(F_2) = (p_*, F'_2)$ are shown in Fig. 1(a,b)). The theorem conditions ensure that resulting sequences have the desired properties.

Cut locus and its computation

The **cut locus** of a point $p \in X$ is the set of all points in X with multiple distinct geodesics to p . The structure of the cut loci in X is vital to the computation of $GC(X)$, as it is exactly the set of points where a GMPR must make a decision.

For X the surface of a polyhedron, there exists a polynomial-time algorithm to compute the cut locus of a given point [3]. Using this algorithm, we obtain the result that the geodesic complexity of the octahedron is four.

Lower bound on $GC(\text{octahedron})$

To show $GC(\text{octahedron}) \geq 4$, we use our simplex embedding theorem on embeddings around a particular point of the cut locus. We construct collections of up to 4-dimensional simplex embeddings to show this lower bound (this is the strongest bound possible with this method). The cut loci relevant to our embeddings are those in Figure 2(c,d,e).

Upper bound on $GC(\text{octahedron})$

We show $GC(\text{octahedron}) \leq 4$ by constructing an explicit geodesic motion planner consisting of 5 GMPRs. We partition the start and end points mainly based on **multiplicity**, the number of unique geodesics between the pair (this strategy has been used for upper bounds on $GC(\text{tetrahedron})$ and $GC(\text{cube})$).

- E_5 is a discrete set containing all (p, q) where p and q are octahedron vertices, or the midpoint of an edge or face (this contains all points with multiplicity 6).
- E_1 (resp. E_4) contains all points not in E_5 with multiplicity 1 (resp. 4).
- E_3 contains all points not in E_5 with multiplicity 3 along with some with multiplicity 2 (which were added to ensure continuity).
- E_2 contains the remaining points with multiplicity 2.

For most of these sets, the associated σ_i is easy to construct. For σ_1 , there is only one choice (since any $(p, q) \in E_1$ has a unique geodesic). E_3 , E_4 , and E_5 can be partitioned into small clopen sets, so σ_3 , σ_4 , and σ_5 are easily constructed as piecewise functions. The most difficult GMPR is σ_2 . To construct $\sigma_2(p, q)$, we consider the cut locus line that q lies on and approach from a counterclockwise direction, when applicable.

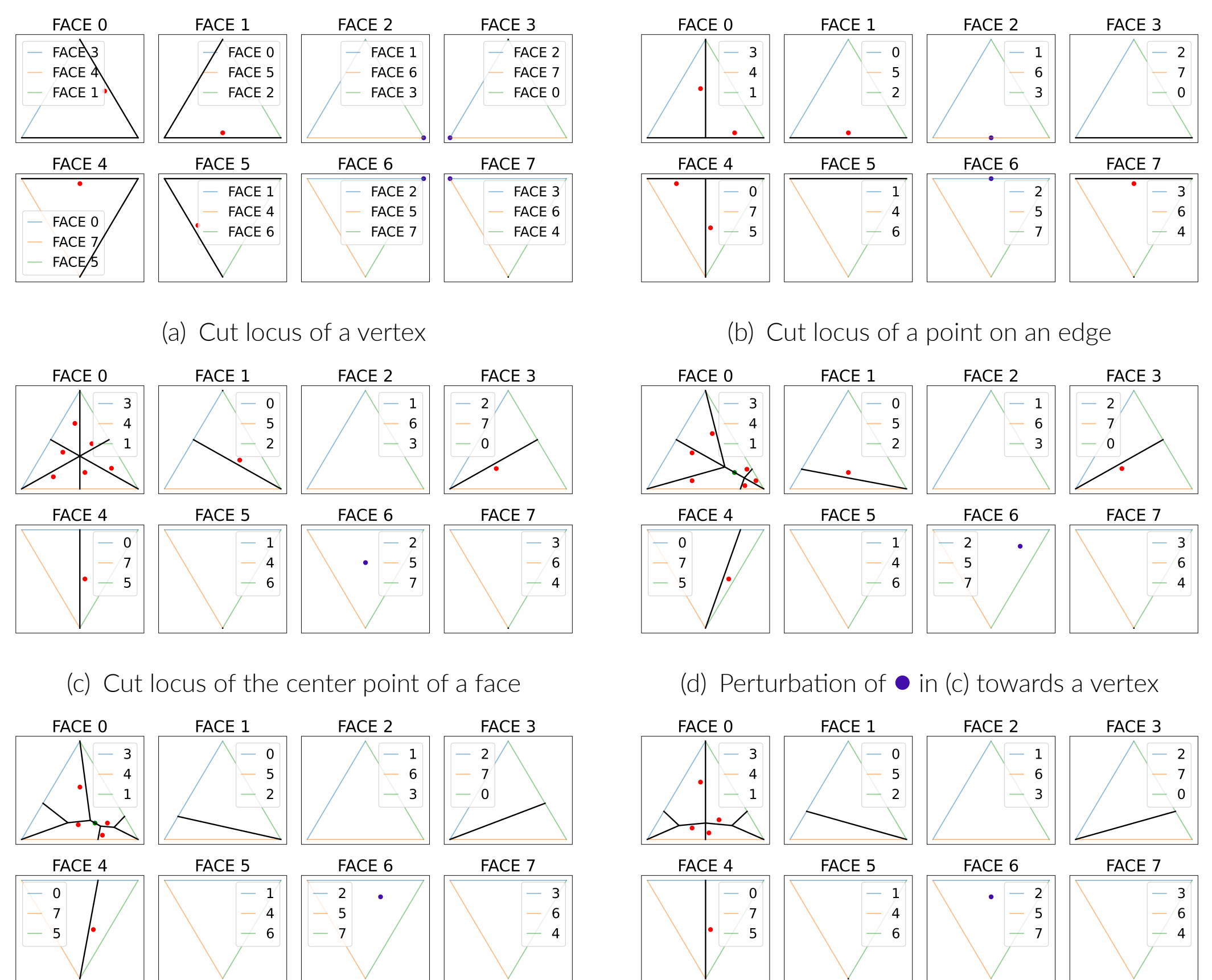


Figure 2. Construction of GMPR σ_2 . A path starting from $\bullet$ approaches a cut locus line from the side indicated by $\bullet$. $\bullet$ indicates free choice on which side to approach from. Observe that when a cut locus line has a unique endpoint on an octahedron vertex, σ_2 approaches the line from the 'right' side of that vertex (i.e. the first side encountered when moving counterclockwise around that vertex).

References

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